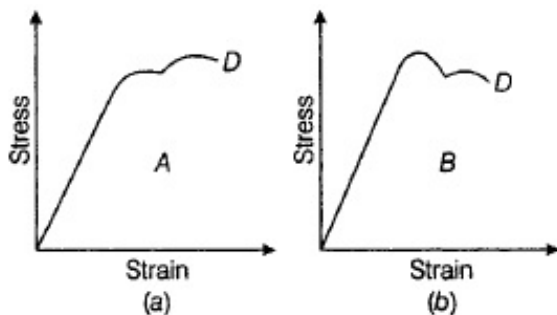

CBSE Test Paper 02
Chapter 9 Mechanical Properties of Solids

1. Columns are loaded in 1
 - a. compression
 - b. tension
 - c. shear
 - d. hydraulic stress
 2. When water freezes, it expands by about 9.00%. What would be the pressure increase inside your automobile's engine block if the water in it froze? (The bulk modulus of ice is $2.00 \times 10^9 \text{ N/m}^2$.) 1
 - a. $2.4 \times 10^8 \text{ N/m}^2$
 - b. $1.1 \times 10^8 \text{ N/m}^2$
 - c. $4.16 \times 10^8 \text{ N/m}^2$
 - d. $1.8 \times 10^8 \text{ N/m}^2$
 3. If the elastic limit of copper is $1.50 \times 10^8 \text{ N/m}^2$, determine the minimum diameter a copper wire can have under a load of 10.0 kg if its elastic limit is not to be exceeded. 1
 - a. 0.812 mm
 - b. 0.912 mm
 - c. 0.712 mm
 - d. 1.012 mm
 4. A steel rod 2.0 m long has a cross-sectional area of 0.30 cm^2 . It is hung by one end from a support, and a 550-kg milling machine is hung from its other end. Determine the resulting strain 1
 - a. 8.0×10^{-4}
 - b. 7.0×10^{-4}
 - c. 9.0×10^{-4}
 - d. 10.0×10^{-4}
 5. A copper cube measures 6.00 cm on each side. The bottom face is held in place by very strong glue to a flat horizontal surface, while a horizontal force F is applied to the upper face parallel to one of the edges. How large must F be to cause the cube to
-

deform by 0.250 mm? (shear modulus of copper = 4.4×10^{10} Pa) **1**

- a. 6.4×10^5 N
- b. 6.5×10^5 N
- c. 6.3×10^5 N
- d. 6.6×10^5 N

6. Show graphically the stress-strain relationship for a brittle material. **1**
7. A wire of length 1 m is stretched so as to make its length 1.01 m on applying an external force. What is the value of strain? **1**
8. What causes variation in velocity of a particle? **1**
9. What is an elastomer? What are their special feature? **2**
10. Two wires made of the same material are subjected to forces in the ratio 1: 4. Their lengths are in the ratio 2: 1 and diameters in the ratio 1: 3. What is the ratio of their extensions? **2**
11. A wire of length L, area of cross-section A and young's modulus Y is stretched by an amount x. What is the work done? **2**
12. Compute the bulk modulus of water from the following data: Initial volume = 100.0 litre, Pressure increase = 100.0 atm ($1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$), Final volume = 100.5 litre. Compare the bulk modulus of water with that of air (at constant temperature). Explain in simple terms why the ratio is so large. **3**
13. The stress-strain graphs for materials A and B are shown in fig,(a) and Fig. (b)



The graphs are drawn to the same scale. **3**

- i. Which of the materials has greater Young's modulus?

ii. Which of the two is the stronger material?

14. The elastic limit of a steel cable is $3.0 \times 10^8 \text{ N/m}^2$ and the cross-section area is 4 cm^2 . Find the maximum upward acceleration that can be given to a 900 kg elevator supported by the cable if the stress is not to exceed one-third of the elastic limit. **3**
15. A steel rod of length $2l$, cross sectional area A and mass M is set rotating in a horizontal plane about an axis passing through the centre. If Y is the Young's modulus for steel, find the extension in the length of the rod. (Assume the rod is uniform.) **5**
-